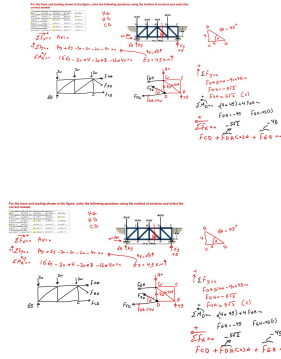
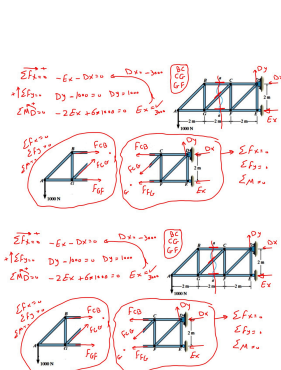
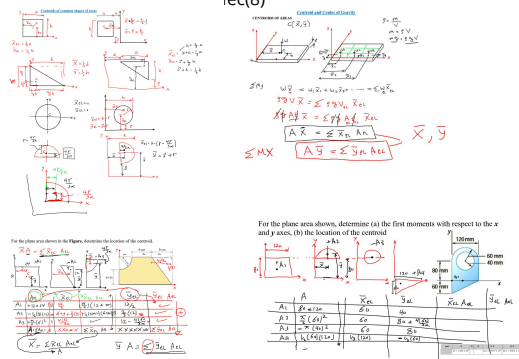


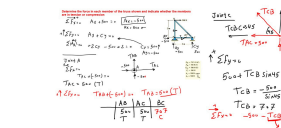
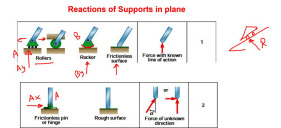
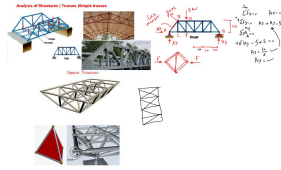
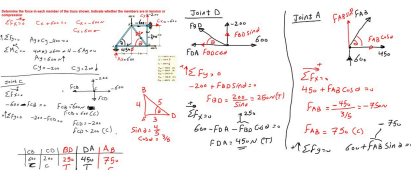
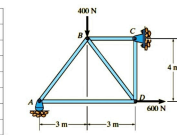
lec(3)

lec(8)



For the truss and loads shown in the figure, solve the following questions and select the correct answer.

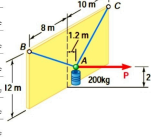
1. The magnitude and direction of the horizontal reaction at point C are:
 - (a) 1000 N, $\rightarrow$
 - (b) 1000 N, $\leftarrow$
 - (c) 1000 N, $\uparrow$
 - (d) 1000 N, $\downarrow$
2. The magnitude and direction of the vertical reaction at point A are:
 - (a) 1000 N, $\uparrow$
 - (b) 1000 N, $\downarrow$
 - (c) 1000 N, $\rightarrow$
 - (d) 1000 N, $\leftarrow$
3. The force in member AB is:
 - (a) 1000 N, $\rightarrow$
 - (b) 1000 N, $\leftarrow$
 - (c) 1000 N, $\uparrow$
 - (d) 1000 N, $\downarrow$
4. The force in member BC is:
 - (a) 1000 N, $\rightarrow$
 - (b) 1000 N, $\leftarrow$
 - (c) 1000 N, $\uparrow$
 - (d) 1000 N, $\downarrow$
5. The force in member CD is:
 - (a) 1000 N, $\rightarrow$
 - (b) 1000 N, $\leftarrow$
 - (c) 1000 N, $\uparrow$
 - (d) 1000 N, $\downarrow$
6. The force in member DE is:
 - (a) 1000 N, $\rightarrow$
 - (b) 1000 N, $\leftarrow$
 - (c) 1000 N, $\uparrow$
 - (d) 1000 N, $\downarrow$
7. The force in member EF is:
 - (a) 1000 N, $\rightarrow$
 - (b) 1000 N, $\leftarrow$
 - (c) 1000 N, $\uparrow$
 - (d) 1000 N, $\downarrow$
8. The force in member FG is:
 - (a) 1000 N, $\rightarrow$
 - (b) 1000 N, $\leftarrow$
 - (c) 1000 N, $\uparrow$
 - (d) 1000 N, $\downarrow$
9. The force in member GH is:
 - (a) 1000 N, $\rightarrow$
 - (b) 1000 N, $\leftarrow$
 - (c) 1000 N, $\uparrow$
 - (d) 1000 N, $\downarrow$
10. The force in member HI is:
 - (a) 1000 N, $\rightarrow$
 - (b) 1000 N, $\leftarrow$
 - (c) 1000 N, $\uparrow$
 - (d) 1000 N, $\downarrow$



Sample Problem

A 200-kg cylinder is hung by means of two cables AB and AC, which are attached to the top of a vertical wall. A horizontal force P perpendicular to the wall holds the cylinder in the position shown. Select the correct answer.

1. The unit vector components $\vec{u}_{AB}$ of the force AB are:
 - (a) $0.866, 0.5, 0$
 - (b) $0.866, 0.5, 0.5$
 - (c) $0.866, 0.5, 0.866$
 - (d) $0.866, 0.5, 0.866$
2. The unit vector components $\vec{u}_{AC}$ of the force AC are:
 - (a) $0.866, 0.5, 0$
 - (b) $0.866, 0.5, 0.5$
 - (c) $0.866, 0.5, 0.866$
 - (d) $0.866, 0.5, 0.866$
3. The unit vector components $\vec{u}_P$ of the force P are:
 - (a) $0.866, 0.5, 0$
 - (b) $0.866, 0.5, 0.5$
 - (c) $0.866, 0.5, 0.866$
 - (d) $0.866, 0.5, 0.866$
4. The force in member AB is:
 - (a) 100 N
 - (b) 120 N
 - (c) 150 N
 - (d) 170 N
 - (e) None of these
5. The force in member AC is:
 - (a) 100 N
 - (b) 120 N
 - (c) 150 N
 - (d) 170 N
 - (e) None of these
6. The force in member AB is:
 - (a) 100 N
 - (b) 120 N
 - (c) 150 N
 - (d) 170 N
 - (e) None of these

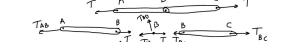


1- The Method of Joints

Types of forces

Tension force (T)

Compression force (C)



lec(7)

REACTIONS AT SUPPORTS

Ball - Roller- Smooth Surface



Ball and socket-Rough Surface

$$\vec{A} = A_x \vec{i} + A_y \vec{j} + A_z \vec{k}$$



Ball and socket-Rough Surface

$$\vec{A} = A_x \vec{i} + A_y \vec{j} + A_z \vec{k}$$

$$\vec{W} = -270 \vec{j}$$

$$\vec{T}_{EC} = \dots$$

$$\vec{T}_{EB} = \dots$$

$$\sum M_A = 0 = (\vec{r}_{AE} \times \vec{T}_{EC}) + (\vec{r}_{AB} \times \vec{T}_{BD}) + (\vec{r}_{AG} \times \vec{W})$$

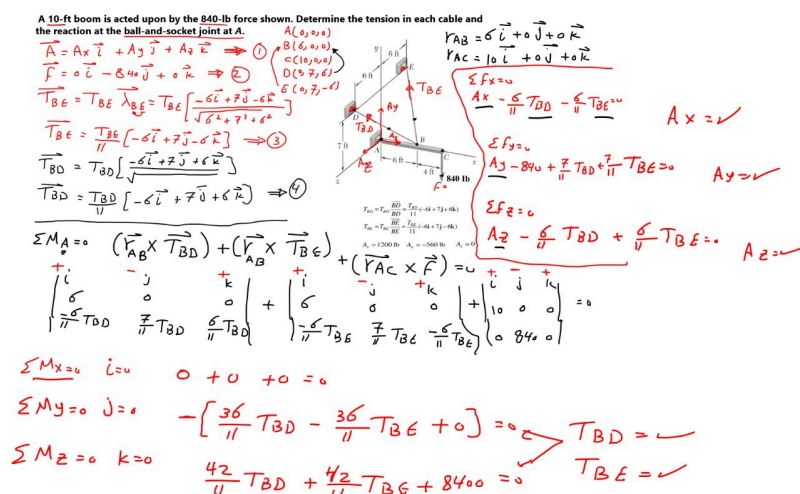
$$\vec{T}_{EC} = \dots$$

$$\vec{T}_{BD} = \dots$$

$$\sum F_x = 0 \quad A_x = \dots$$

$$\sum F_y = 0 \quad A_y = \dots$$

$$\sum F_z = 0 \quad A_z = \dots$$



A 5 x 8-m plate of uniform density weighs 270 N and is supported by a ball-and-socket joint at A and by two cables. Determine the tension in each cable and the reaction at A.

